Presenter Script — Standardization of Spirometry: ATS/ERS 2019 Update

Speaker: Sasanka Barman
Reference: Am J Respir Crit Care Med 200:e70–e88 (2019)

Slide 1 — Title Slide

Good [morning/afternoon], everyone. Today I'll be walking through the key technical standards and clinical implications from the **2019 ATS/ERS update on spirometry standardization**. This is a critical document not just for pulmonologists and respiratory therapists, but also — and this is something I'll circle back to — for embedded firmware developers working on spirometry hardware like the **MSP430FR6043 ultrasonic flow measurement system**. Let's dive in.

Slide 2 — Agenda

Before we get into the details, let me give you a quick map of where we're going.

Spirometry is **the most common pulmonary function test** performed worldwide. Accurate, reproducible measurements are foundational — they drive diagnosis, disease monitoring, and treatment decisions. The 2019 update didn't just tweak a few numbers; it **revised every major criterion** — from how we judge a maneuver acceptable, to equipment accuracy requirements, to the grading system itself.

We're going to cover eight areas today: Introduction and scope, equipment and QA, the FVC/FEV_1 maneuver, acceptability criteria, key parameters, the grading system, bronchodilator testing, and finally, what all this means for device design. The update drew on **over 190 directly relevant studies** and input from more than **1,760 patients surveyed** — so this isn't small revisions; it's a comprehensive evidence-based overhaul.

Slide 3 — Introduction & Scope

Let's start with the basics. Spirometry is defined as a physiological test measuring the **maximal volume of air an individual can inspire and expire with maximal effort**, where the primary signal is either flow or volume as a function of time.

The primary variables we report are:

- **FVC** — Forced Vital Capacity: total volume expelled from full inspiration with maximal effort

- **FEV_1** — Forced Expiratory Volume in the first second
- **FEV_1/FVC ratio** — our key indicator of airflow obstruction
- **PEF** — Peak Expiratory Flow
- **$FEF_{25–75}$** — Mean flow in the mid-expiratory phase, which is sensitive to small airway disease

The 2019 document covers all of these across six major domains: definitions and terminology, equipment specs per ISO 26782, patient preparation and contraindications, maneuver acceptability and repeatability, the quality grading system, and data reporting standards. Everything is interconnected, so a change in one area tends to cascade through the others.

Slide 4 — Equipment & Quality Assurance

Now, no measurement is better than the device taking it, so let's talk about equipment requirements and QA.

The standard requires **ISO 26782 compliance**, with a maximum permissible error of **±2.5%**. Key hardware specs include: flow range of **0–14 L/s**, volume capacity of **≥8 L** at BTPS conditions, sampling rate **>100 Hz**, minimum **12-bit ADC resolution**, and a resistance of **<1.5 $cmH_2O·L^{-1}·s^{-1}$**.

For **daily QA**, the routine is: verify with a **3-L syringe** — tolerance is ±3%, or more precisely ±0.015 L or ±0.5% of full scale. Set zero-flow baseline before each session, ensure any filter used in testing was also used during calibration, and log all calibration results and failures. Monthly leak tests are required, and any syringe that has been **dropped or damaged is out of calibration until verified**.

A point I want to highlight here: **BTPS correction** — Body Temperature, Pressure, Saturated. All results must be reported under BTPS conditions. If you assume ambient temperature instead of measuring it, you can introduce errors up to **6%**. For volume spirometers, temperature should be measured per breath, and operators should be alerted if temperature changes by more than **3°C within 30 minutes**.

Slide 5 — FVC / FEV_1 Maneuver — Four Phases

The FVC/FEV_1 maneuver has four distinct phases, and each one has specific technical requirements.

Phase 1 — Maximal Inspiration: The patient inspires rapidly to total lung capacity (TLC). The pause at TLC must be **less than 2 seconds** — a slow or prolonged pause will artificially reduce both PEF and FEV_1.

Phase 2 — "Blast" of Expiration: Immediate maximal forced expiratory effort. This is where Back-Extrapolated Volume (BEV) is determined — it must be **<5% of FVC or 0.100 L**, whichever is smaller. PEF rise time — from 10% to 90% — must be **<150 milliseconds**.

Phase 3 — Continued Full Expiration: The patient sustains the effort for up to **15 seconds**. The end-of-forced-expiration (EOFE) plateau criterion is less than **0.025 L change in 1 second**. Devices should provide audio feedback: one beep signals the plateau has been reached; two beeps signal the 15-second time limit.

Phase 4 — Maximal Inspiration (Post-EOFE): The patient rapidly returns to full inspiration, generating the FIVC — Forced Inspiratory Vital Capacity — which is used to confirm the starting volume. If **FIVC minus FVC exceeds 0.100 L**, the maneuver is considered invalid because it suggests the patient didn't start from a true full inspiration.

Slide 6 — Within-Maneuver Acceptability Criteria

This is one of the most practically important slides. The 2019 update distinguishes between **acceptable** and **usable** maneuvers — a key distinction from prior standards.

For **FEV_1 acceptability**, the maneuver must meet: BEV <5% FVC or 0.100 L, no faulty zero-flow setting, no cough in the first second, no glottic closure in the first second, no obstructed mouthpiece, no leak, and FIVC − FVC <0.100 L.

For **FVC acceptability**, the criteria also include: no glottic closure after 1 second, and the EOFE plateau criterion of <0.025 L in 1 second.

Usable FEV_1 maneuvers relax some criteria — notably, a maneuver can be usable even if the mouthpiece was obstructed or there was a leak — as long as the first second is clean.

Think of this table as a checklist your device or technician needs to run through for every single blow.

Slide 7 — EOFE Criteria & Between-Maneuver Repeatability

For end of forced expiration, **any one of three criteria** is sufficient:

1. The EOFE **plateau**: <0.025 L change for at least 1 second — triggers a single beep
2. **Time limit**: Forced Expiratory Time ⩾ 15 seconds — triggers a double beep
3. **Restrictive pattern**: FVC equals or exceeds the prior acceptable FVC — applies particularly in children and patients with restrictive disease

For **between-maneuver repeatability**, in patients over 6 years, the two largest FVC values must be within **0.150 L** and the two largest FEV_1 values within **0.150 L**. Children ≤6 years have tighter criteria — within 0.100 L or 10%, whichever is greater.

The goal is a minimum of **3 acceptable maneuvers** for both FEV_1 and FVC — though they don't have to come from the same maneuver. Adults are capped at **8 attempts** in normal circumstances. And critically — if at any point an acceptable FEV_1 drops below **80% of the starting value**, testing should be terminated for safety.

Slide 8 — Quality Grading System

The 2019 update formalized a grading system from A through F, plus U, applied **separately** for FEV_1 and FVC, and separately for pre- and post-bronchodilator sessions.

- **Grade A**: ≥3 acceptable maneuvers, repeatability <0.150 L — highest confidence the result is the patient's true best
- **Grade B**: 2 acceptable, repeatability <0.150 L
- **Grade C**: ≥2 acceptable, repeatability <0.200 L
- **Grade D**: ≥2 acceptable, repeatability <0.250 L
- **Grade E**: ≥2 acceptable, repeatability >0.250 L, or only 1 acceptable maneuver
- **Grade U**: 0 acceptable but ≥1 usable maneuver
- **Grade F**: 0 acceptable and 0 usable

Important nuance: grades below A are **still clinically informative**. The grade reflects interpreter confidence in the result, not a verdict on lung health. For children ≤6 years, the thresholds tighten — the criterion becomes 0.100 L or 10% of the highest value, whichever is greater.

Slide 9 — Bronchodilator Responsiveness Testing

Bronchodilator testing follows a four-step protocol: baseline spirometry, bronchodilator administration, a wait period, then post-BD spirometry — with **≥3 acceptable maneuvers** required at both ends.

The dose and route follow facility protocol, but **withholding times are standardized by agent class**:

- Short-acting beta-agonists (SABAs) like albuterol: **4–6 hours**
- Short-acting muscarinic antagonists (SAMAs) like ipratropium: **12 hours**
- LABAs like formoterol or salmeterol: **24 hours**

- Ultra-LABAs like indacaterol or vilanterol: **36 hours**
- LAMAs like tiotropium or umeclidinium: **36–48 hours**

For reported values, we report **FVC and FEV_1** to two decimal places at BTPS; **FEV_1/FVC** as a decimal fraction to 0.01; PEF in L/s; FET in seconds; FIVC; and critically — **delta FEV_1 and delta FVC as both absolute change and percentage change** versus pre-BD. FEF_{25-75} is reported from the maneuver with the **largest combined FEV_1 + FVC sum**.

Slide 10 — Device Design Implications

For those of us on the device development side, every clinical requirement in this standard maps directly to a firmware requirement.

Signal acquisition: >100 Hz sampling, ⩾12-bit ADC, full flow range 0–14 L/s, volume 0.5–8 L BTPS, back-extrapolation to compute Time Zero, and BEV detection and flagging.

Real-time processing: Simultaneous flow-volume and volume-time display, PEF rise time detection with the <150 ms threshold, EOFE plateau detection at <0.025 L per second, audio alert generation for both EOFE conditions, FET timer with a hard 15-second stop, and per-breath BTPS compensation.

QA and reporting: Automated acceptability feedback to the operator, full Grade A–F and U flagging, calibration log generation, ±2 SD alerting on calibration factor drift, zero-flow verification before each test, and operator override capability.

In the context of the **MSP430FR6043**, the USS software library computes AbsTOF for both upstream and downstream flow paths at 400 Hz — all six of these firmware requirements map directly to that pipeline design.

Slide 11 — Key Takeaways

Let me close with the six things I want you to walk away with:

1. **Equipment**: ISO 26782 compliance, ±2.5% max error, >100 Hz sampling, daily 3-L syringe verification.
2. **Maneuver**: Four phases, BEV <0.100 L, EOFE plateau <0.025 L in 1 second, back-extrapolation to define Time Zero.
3. **Acceptability**: Nine within-maneuver criteria, minimum 3 acceptable maneuvers each for FEV_1 and FVC, capped at 8 adult attempts.
4. **Grading A–F and U**: Grade A = highest confidence. Grading informs interpreter confidence — not a statement about lung health. Thresholds tighten for younger patients.

5. **Bronchodilator Testing**: Pre- and post-BD spirometry with agent-specific withholding times from 4 hours up to 48 hours; always report both absolute and percent change.
6. **Device Design**: >100 Hz ADC, real-time BTPS, plateau detection, dual-display output, and automated operator feedback are non-negotiable requirements.

Thank you. I'm happy to take questions.

References: ATS/ERS 2019 — Graham et al., Am J Respir Crit Care Med 200:e70–e88 (2019); Miller et al., Eur Respir J (2005)